

Erratum: Equation of State for Solid Phase I of Carbon Dioxide Valid for Temperatures up to 800 K and Pressures up to 12 GPa [J. Phys. Chem. Ref. Data 40, 043105 (2011)]

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Key words: CO₂; compressibility; density; equation of state; expansivity; heat capacity; Helmholtz energy; melting curve; molar volume; solid carbon dioxide; sublimation curve.

Three corrections are required to the original article,¹ as detailed below.

- (1) In Eq. (38), a factor $\theta_{D,0}$ was missing; the correct form of the equation is as follows:

$$A_A = b_1 R \theta_{D,0} [(T/\theta_{D,0})^4 / \{1 + b_2 (T/\theta_{D,0})^2\}] \exp[b_3 \{(V_m - V_{00})/V_{00}\}]. \quad (38)$$

- (2) The parameters appearing in Eq. (49) were given in the paragraph following that equation with the wrong signs; the correct values are $d_7 = -0.160433$, $d_8 = 0.018643$, and $d_9 = -0.001582$.
- (3) In Table 6, the value of parameter b_1 was given incorrectly as 1.15; the correct value is 1.151. There were also several typographical errors in the symbols appearing in Table 6. The corrected table is reproduced below in full.

A software implementation of the equation of state is available from the author.

TABLE 6. Parameters of the new equation of state.

Parameter	Value	Parameter	Value
$V_{00}/(\text{cm}^3 \text{ mol}^{-1})$	25.800	b_1	1.151
c_1/MPa	9000	b_2	55.428
c_2/MPa	32315	b_3	8.8
c_3/MPa	941	$\theta_{D,0}/\text{K}$	148.9
$\theta_{1,0}/\text{K}$	92.3	a_1	0.5
$\theta_{2,0}/\text{K}$	179.4	a_2	1.5
$\theta_{3,0}/\text{K}$	1945.9	a_3	1
$\theta_{4,0}/\text{K}$	942.7	a_4	2
$\theta_{5,0}/\text{K}$	3379.2	a_5	1
$\gamma_{D,0}$	2.6163	q_D	-0.21
$\gamma_{1,0}$	4.0463	q_1	-0.21
$\gamma_{2,0}$	1.1259	q_2	-0.21
$\gamma_{3,0}$	0.0633	q_3	0
$\gamma_{4,0}$	-0.0521	q_4	0
$\gamma_{5,0}$	0.0515	q_5	0

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References

- ¹J. P. M. Trusler, J. Phys. Chem. Ref. Data **40**, 043105 (2011).

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